# CappedMinterV3 Review

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Prepared for ZKsync

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## About the ZKsync CappedMinterV3 and ZkMinterTrigger Review

ZKsync Era is a Layer 2 ZK rollup that uses cryptographic validity proofs to provide scalable and low-cost transactions on Ethereum. The CappedMinterV3 contract is an updated version of the token minting system that allows the owner to mint up to a preselected amount of tokens with enhanced functionality including updateable mintable addresses and support for multiple default admins.

## About Offbeat Security

Offbeat Security is a boutique security company providing unique security solutions for complex and novel crypto projects. Our mission is to elevate the blockchain security landscape through invention and collaboration.

## Summary & Scope

The src folder was reviewed at commit aa60218.

The following **2 files** were in scope:

- src/ZkCappedMinterV3.sol
- src/ZkCappedMinterV3Factory.sol

## Summary of Findings

| Identifier | Title | Severity | Fixed |
|---|---|---|---|
| I-01 | Admin-address-sanity-check | Informational | PR#46 |

## Additional Recommendations

**Note**: The `ZkCappedMinterV3Factory` contract accepts a bytecode hash as a constructor parameter without verifying it matches the actual deployment bytecode of `ZkCappedMinterV3`. This could lead to incorrect address predictions by the `getMinter()` function if the provided hash doesn't match the actual deployed contract bytecode. This issue been reported on previous versions which the project has acknowledged and stated they will take care to deploy the factory with the correct bytecode hash.

## Detailed Findings

## Informational Findings

### [I-01] Admin address sanity check

#### Summary

The ZkCappedMinterV3 constructor and factory deployment methods lack validation to ensure the admin address is non-zero. This allows deployment of minter instances with administrative roles granted to address(0), resulting in permanently unmanageable contracts that cannot mint tokens or be administered.

#### Description

The ZkCappedMinterV3 constructor accepts an `_admin` parameter and grants it both `DEFAULT_ADMIN_ROLE` and `PAUSER_ROLE` without validating that the address is non-zero:

```
constructor(IMintable _mintable, address _admin, uint256 _cap, uint48 _startTime,
    // ... time validation ...

    CAP = _cap;
    START_TIME = _startTime;
```

```
    EXPIRATION_TIME = _expirationTime;

    _updateMintable(_mintable);
    _grantRole(DEFAULT_ADMIN_ROLE, _admin);  // No validation that _admin != add
    _grantRole(PAUSER_ROLE, _admin);
}
```

### Recommendation

Consider adding validation in the ZkCappedMinterV3 constructor to ensure the admin address is non-zero:

```
constructor(IMintable _mintable, address _admin, uint256 _cap, uint48 _startTime,
+   if (_admin == address(0)) revert ZkCappedMinterV3__InvalidAdmin();
    if (_startTime > _expirationTime) {
        revert ZkCappedMinterV3__InvalidTime();
    }
```